

Data Mining – Lecture 11 – Outlier Analysis IS422P

1-Outlier is a data object that deviates significantly from the normal objects.

- a- True
- b- False

Answer : a.

2-Outlier and noise data are quite the same .

- a- True
- b- False

Answer : b.

3- Temperature value is an example of ____ outlier .

- a- Global
- b- Contextual
- c- Collective
- d- None

Answer : b.

4- _____ is a challenge for outlier detection.

- a- Modeling normal objects
- b- Handling noise
- c- Understandability
- d- All above

Answer : d.

5- Statistical , proximity based and clustering based methods are based on whether user labeled examples of outliers can be obtained.

- a- True
- b- False

Answer : b, explanation : “ based on assumptions about normal data and outliers “

6-What is the main approach of supervised methods for outlier detection?

- a) Modeling outlier detection as a clustering problem
- b) Modeling outlier detection as a regression problem
- c) Modeling outlier detection as a classification problem
- d) Modeling outlier detection as an optimization problem

Answer : c.

7- In supervised outlier detection methods, what is often boosted to handle imbalanced classes?

- a) Normal class
- b) Outlier class
- c) Both classes equally
- d) No class is boosted

Answer : b.

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8-Recall is more important than accuracy in supervised outlier detection methods to ensure all outliers are detected.

- a- True
- b- False

Answer : a.

9- Which statement is true for unsupervised outlier detection methods?

- a- Normal objects are randomly distributed without any pattern
- b- Outliers are expected to be close to groups of normal objects
- c- Normal objects are somewhat clustered into multiple groups
- d- Outliers and normal objects share strong patterns

Answer : c.

10-What is a significant weakness of unsupervised outlier detection methods?

- a- High accuracy in detecting all outliers
- b- Inability to detect individual outliers effectively
- c- High false positive rate and missing many real outliers
- d- Low false positive rate but missing many normal objects

Answer : c.

11-In semi-supervised outlier detection methods, what is used if some labeled normal objects are available?

- a- Use the labeled examples and the proximate unlabeled objects to train a model for normal objects
- b- Use only labeled outliers to train the model
- c- Ignore the labeled data and use unsupervised methods
- d- Use labeled examples to model outliers

Answer : a.

12- Supervised methods are considered more effective in identifying attacks on key resources.

- a- True
- b- False

Answer : a.

13- What is a challenge faced by semi-supervised outlier detection methods when only labeled outliers are available?

- a- The labeled outliers may not represent all possible outliers
- b- The labeled normal objects may dominate the model
- c- The model may become too complex
- d- The method cannot use unlabeled data

Answer : a.

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14-What approach can improve the quality of detection in semi-supervised methods when labeled outliers are limited?

- a- Ignoring labeled outliers
- b- Using models for normal objects learned from unsupervised methods
- c- Increasing the number of labeled outliers
- d- Using only supervised learning techniques

Answer : b.

15- Model based methods is another name for statistical methods in outlier detection.

- a- True
- b- False

Answer : a.

16-How do statistical methods identify outliers?

- a- By clustering data points into groups
- b- By measuring the distance between data points
- c- By comparing data against a statistical model
- d- By evaluating the density of data regions

Answer : c.

17- On what does the effectiveness of statistical methods for outlier detection primarily depend?

- a- The size of the dataset
- b- The accuracy of clustering algorithms
- c- The assumption of the statistical model holding true in real data
- d- The complexity of the model used

Answer : c.

18-Which of the following describes parametric statistical methods?

- a- They assume that normal data follows a specific distribution with parameters
- b- They do not make any assumptions about the distribution of normal data
- c- They learn the model from the input data using histograms
- d- They use clustering to identify normal data

Answer : a.

19- Histogram is an example of non-parametric method for outlier detection.

- a- True
- b- False

Answer : a.

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20- What is the key distinction between parametric and non-parametric statistical methods?

- a- Parametric methods assume a specific data distribution, while non-parametric methods learn the model from input data
- b- Parametric methods require labeled data, while non-parametric methods do not
- c- Parametric methods use clustering, while non-parametric methods use distance metrics
- d- Parametric methods are used for classification, while non-parametric methods are used for regression

Answer : a.

21- In cluster based outlier detection methods , normal data are typically characterized as belonging to a large and dense clusters .

- a- True
- b- False

Answer : a.

22- Which of the following is NOT a criterion for detecting outliers in clustering-based methods?

- a- Object does not belong to any cluster
- b- Object belongs to a small and remote cluster
- c- Object has a large distance to the closest cluster
- d- Object is part of the largest cluster

Answer : d , note “ this is based on our slides , idk if it’s possible “

23- LOF – Local outlier factor is computed in terms of the KNN of an object in comparison to its neighbors .

- a- True
- b- False

Answer : a .

24-What is a key disadvantage of classification-based methods for outlier detection?

- a- They require clustering of data points
- b- They are slow and inefficient for large datasets
- c- They are heavily biased due to the imbalance in the training set
- d- They cannot handle multi-class classification problems

Answer : c.

25- How does a one-class model work in classification-based outlier detection?

- a- It uses clustering to group normal and outlier data points
- b- It describes only the normal class and detects outliers as those not within the normal class boundary
- c- It uses multiple classes to describe normal behavior
- d- It combines supervised and unsupervised learning methods

Answer: b.

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26-Detecting new outliers that weren't presented in the training set is an advantage of using a one-class model in outlier detection.

- a- True
- b- False

Answer : a.

27-What is a major bottleneck of semi-supervised learning methods for outlier detection?

- a- High computational cost
- b- Low detection speed
- c- Dependence on the availability and quality of the training set
- d- Inability to detect collective outliers

Answer: c.

“For the following nine points (2,2), (2,7), (3,1), (3,5), (4,3), (4,8), (5,2), (6,2), (6,5)

☐ Assume that $k = 2$ and initially assign (2,2) and (2,7) as the center of each cluster.

☐ Apply the k-means algorithm using the Manhattan distance and show the new cluster centers after the first round execution (Hint: The Manhattan distance is: $d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{in} - x_{jn}|$.) “

28- Given the initial cluster centers (2,2) and (2,7), which points will belong to Cluster 1 after the first round of k-means execution using Manhattan distance?

- a- (2,7), (3,5), (4,8), (6,5)
- b- (2,2), (3,1), (4,3), (5,2), (6,2)
- c- (2,2), (3,1), (3,5), (5,2)
- d- (4,3), (5,2), (6,2)

Answer: b.

29-Which point has the shortest Manhattan distance to the initial cluster center (2,7)?

- a- (2,2)
- b- (3,1)
- c- (3,5)
- d- (4,3)

Answer : c.

30- The new mean for Cluster 1 is ____ .

- a- (2,2)
- b- (3,2)
- c- (3,5)
- d- (4,2)

Answer : d.

" شكرا لكم ولا تنسونا بالدعاء "